



Security Watch

Update Your JRE And Adobe Reader Now!

A vulnerability has been reported in the Sun Java Runtime Environment (JRE). It is caused due to an error when processing GIF images; it can be exploited via specially-crafted GIF images. Successful exploitation can mean the taking over of your computer by Those With Malicious Intent (TWMI). The vulnerability is reported in JDK with JRE 5.0 Update 9 and prior; SDK with JRE 1.4.2_12 and prior; and SDK and JRE 1.3.1_18 and prior. Security firm Secunia rates the vulnerability "Highly Critical." In all probability, you downloaded Java for the JRE at some point of time. The JRE allows you to run Java applications off the Net. To see what version of the JRE you have, go to the Java applet in the Control Panel, and in the General tab, click About.

Strictly speaking, there's no need to check what version you have—simply upgrade to the latest version. Head to <http://java.sun.com/j2se/1.4.2/download.html> to download the J2SE JRE, version 1.4.2_13.

A vulnerability has also been reported in Adobe Reader. It is caused due to an unspecified error when processing PDF files; it can be exploited via a specially-crafted PDF file. Again, Successful exploitation can mean the taking over of your computer by TWMI.

The vulnerability is reported in version 7.0.8 and prior, though other versions may be affected, too. Fixing this is a simple matter of upgrading to the latest version—8.0—which you'd want to do anyway, even without the vulnerability. At www.adobe.com, click "Get Adobe Reader."

news in chip design has been all about in recent years. Reversible computing and DNA computing and other "computings" are being heavily researched, as are solutions such as using nanotubes in transistors.

HP has once again validated its claim to its own tagline, "Invent," with research that proves it is possible to do away with copper interconnects in chips, by using grids of nanowires. See figure below: it shows the structure of HP's proposed "crossbar" network of nanowires. Two layers of crossbars with orientations perpendicular to each other form a sandwich outside a layer of silicon transistors. The transistors and other functional blocks in the chip communicate through "dynamic" (meaning non-hardwired) connections created in the crossbars.

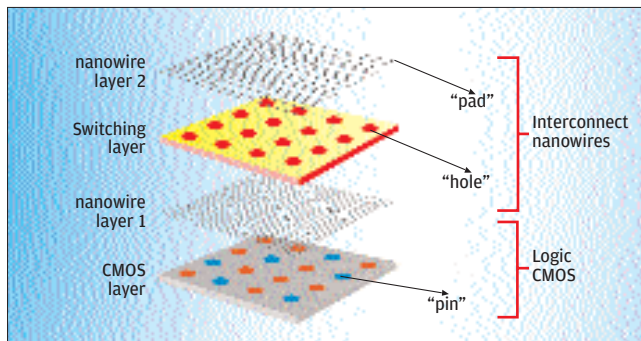
Think of a nanowire as just a wire on the nanoscale—specifically, with a width in the range of tens of nanometres. Because of their extreme thinness, they are often referred to as one-dimensional structures.

The new design will achieve three objectives: reducing size, increasing performance, and reducing power consumption.

It should be obvious that chips will shrink drastically once the interconnects are gotten rid of. The beauty of the crossbar method is that traditional transistors can still be used.

The company has thus far created a simulation of a field-programmable gate array (FPGA) with a crossbar grid, and hopes to have a prototype by the end of the year. (An FPGA is a type of programmable chip; the simulation essentially maps the intended circuit to the nanowire structure.) The timeframe for commercial chips using the crossbar grid structure is 2010—HP says it is possible for the concept to be incorporated into chips of other types, though the simulation only demonstrated it for FPGAs.

With the FPGA as an example, the functional blocks are hardwired to each other through interconnects, so an



HOT

The iPhone



It's so cool! There's no keypad, a multiple-touch screen, up to 8 GB of space, it

reorients the screen depending the way you're holding it, has Wi-fi and did we mention how cool it looks? It also makes phone calls.

The iPhone

NOT



You can't change the battery, there's no 3G, it won't run just *any* application like your phones do today, the screen isn't purely scratch-resistant, it's grossly overpriced and is available only with Cingular today. We'll wait for a cheaper me-too, thank you.

increase in the number of functional blocks leads to a "geometric" increase in the number of data pathways. Now, the pathways themselves could consume 80 per cent of available chip area. The crossbar system would result in an "intelligent communications system" that would only connect functional blocks when they need to be connected.

In other chips, with a dynamic network performing the role of interconnects, some transistor areas could be turned off when not in use, thus saving on power.

According to HP's estimates, an FPGA made with 45-nm transistors and a grid of nanowires 4.5 nanometres across would



One Silly Question

"What Valentine gift would you give your computer?"



"A new game"
Abdullah Manihar



"An Anti-Virus"
Ambarish Patil



"e-flowers"
Priyanka Babu



"UPS for load shedding"
Pratik Desai



"A good mouse"
Sarita Balakrishnan